

	$= 8.14 \text{ cm}$
4(a)	When the gas particles approach the walls of the vessel, they collide with the wall. The wall exerts a force on the particles that causes them to change in momentum according to Newton's second law of motion. By Newton's third law of motion, the particles exert a reaction force on the wall, and hence the gas exerts pressure on the walls.
4(b)(i)	There are no intermolecular forces between the particles, because they are spaced very far apart.
4(b)(ii)	$pV = NkT$ $N = \frac{pV}{kT} = \frac{(180)(3.2 \times 10^{-4})}{(1.38 \times 10^{-23})(298)} = 1.40 \times 10^{19}$ Assuming that the vessel is a cube, $\text{estimated average distance} = \frac{\text{length of edge of vessel}}{\text{number of particles along each edge}}$ $= \frac{\sqrt[3]{3.2 \times 10^{-4}}}{\sqrt[3]{1.40 \times 10^{19}}}$ $= 2.84 \times 10^{-8} \text{ m}$
4(b)(iii)	The diameter of the particle is much smaller than the average distance in (b)(ii) , hence the assumption in (b)(i) is valid.
5(a)	1. There is no net flow of energy in a stationary wave